Getting Started with CI/CD AWS Native Way

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Why This Session Exists

- Writing code and shipping code are different disciplines. Most engineering failures happen not in the code itself, but in how it reaches production manually, inconsistently, and without guardrails.
- **CI/CD is the industry standard, not a preference.** Every production engineering team operates inside a pipeline. Automated build, test, and deployment cycles are not optional practices; they are the baseline expectation for any software role at scale.
- **Automation eliminates a class of errors.** Manual deployments introduce human variance. A pipeline enforces the same steps, in the same order, every time, regardless of who pushed the commit or what day it is.
- **AWS native tooling is what you will encounter.** CodePipeline, CodeBuild, and ECS are not academic tools. They are what engineering teams at companies running on AWS use in production today. Starting with them means starting with what's real.

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What is CICD

A discipline automated by tools that turns a code push into a running application

01 **Automated.**
No manual build, no manual deploy, no SSH into a server.

02 **Repeatable.**
The same steps run every single time, in the same order, with the same result.

03 **Fast feedback.**
Broken code is caught in minutes, not in production at 4 pm on a Friday.

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CI vs CD vs CD.

Three letters, three ideas often confused, rarely explained.

CI

Continuous Integration

Every push is automatically built and tested. Nobody merges broken code.

CD

Continuous Delivery

Every green build is ready to deploy. A human clicks the release button.

CD

Continuous Deployment

Every green build deploys automatically. No human in the loop.

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The pipeline mental model

Most CICD pipeline - anywhere , any tool - looks like this

Stages are connected by artifacts — the output of one stage becomes the input of the next.

`Pipeline`	The end-to-end automation.
`Stage`	One step in the automation does one job, hands off.
`Artifact`	What moves between stages — a zip, an image, a config file.

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DevOps tools: a full set of tools.

The full developer tools lifecycle — author, source, build, deploy, release, monitor.

AUTHOR	SOURCE / ARTIFACT	BUILD & TEST	DEPLOY	RELEASE	MONITOR
Kiro AWS Toolkits & SDKs	AWS CodeConnections AWS CodeArtifact AWS CodeCommit	AWS CodeBuild	AWS CodeDeploy	AWS AppConfig	AWS X-Ray Amazon CloudWatch Amazon DevOps Guru AWS Config Managed Grafana Managed Prometheus

ORCHESTRATE AWS CodePipeline

MODEL AWS CloudFormation · AWS CDK · AWS SAM

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Meet the services.

Six AWS services — each with one job — come together to form the pipeline.

CodePipeline

Orchestrates the whole flow

CodeBuild

Compiles, tests, packages

CodeDeploy

Zero-downtime deploys

ECR

Stores Docker images

ECS

Runs the containers

IAM

IAM - Role over long-live creds

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CodePipeline — the orchestrator

If you remember one service from today, make it this one.

01

It doesn't build. Its wires up the Actions that Do .

CodePipeline coordinates every other service does the actual work.

02

It watches your source.

A commit to GitHub triggers the whole pipeline, automatically.

03

It gives you a visual map.

Every stage lights up green or red. You see exactly where things stand.

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Supporting cast.

Each service does exactly one thing — that's the point.

CodeBuild

Managed build server. No infra to patch. Runs your buildspec, streams logs, shuts down.

CodeDeploy

Deployment specialist. Blue/Green, rolling, and canary strategies with automatic rollback.

ECR

Private Docker registry. Every image tagged with a commit SHA — fully traceable.

ECS

Serverless container runtime on Fargate. Health-check gated. Pay per task-second.

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Pain points: how AWS answers them.

These problems don't disappear on AWS. But AWS native tools are built for them.

THE PAIN	AWS NATIVE ANSWER
Keeping up with best practices. Pipeline YAML grows. Every team reinvents.	**CDK Pipelines + CodePipeline V2.** Opinionated, versioned templates. Pipeline-level variables.
Ensuring security. Hardcoded secrets, overprivileged roles, no audit trail.	**IAM roles per service + Secrets Manager.** No passwords. Least privilege. All actions logged to CloudTrail.
Scaling as you grow. Jenkins master becomes a bottleneck during release windows.	**CodeBuild is serverless.** Scales to parallel builds with zero config.
Troubleshooting & maintenance. “Why did this fail at 2am?” Logs are scattered.	**CloudWatch + AWS DevOps Agent.** Every run logged. New DevOps Agent correlates root cause.
Managing cost. Build servers run 24/7 whether building or not.	**Pay per build-minute, per task-second.** Fargate charges only while your container is up. Zero idle.

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Shift left — and why every time.

The later you find a problem, the more it costs. Security belongs in the commit, not in the firewall.

THE COST CURVE

COMMIT	PULL REQUEST	STAGING	PRODUCTION
Minutes	**Hours**	**Days**	**Weeks**
Dev catches it locally	Reviewer flags the pattern	QA cycle + redeploy	Incident + postmortem + comms

WHAT TO SHIFT LEFT — AND WHERE

AT COMMIT

Pre-commit hooks + secret scanning

git-secrets, trufflehog. Blocks credentials before they reach the repo.

AT BUILD

SAST + dependency + image scanning

Trivy, Snyk, Amazon Inspector in a CodeBuild phase. Fail on critical CVEs.

AT DEPLOY

Policy + config checks

Least-privilege IAM, non-root containers, no public S3 — all verified pre-deploy.

IN PROD

Runtime detection — not a replacement

CloudWatch, GuardDuty. The safety net, not the first line.

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Best practices — always apply.

Seven habits that separate a demo pipeline from a production one.

01 **Never commit secrets.**

Use Secrets Manager or SSM Parameter Store. Reference by name in buildspec — never hardcode.

02 **Pin your base images.**

FROM node:20-alpine is fine. FROM node:latest is a liability.

03 **Tests run before the build.**

If your tests run after Docker build, you're building untested code. Move to pre_build. Fail fast.

04 **Every image gets a commit SHA tag.**

‘latest’ is a lie. Tag with Git SHA — you can trace a container back to an exact line of code.

05 **Health checks are not optional.**

ECS won't know your container is broken without one. Every service needs a /health endpoint.

06 **Fail the pipeline — don't warn.**

A scan that warns but doesn't fail is decoration. A critical CVE must stop the pipeline.

07 **Pipeline is the only deploy path.**

Nobody SSHs into prod. If you need to bypass it, the pipeline is broken — fix the pipeline.

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The future is agentic — re:Invent 2025.

Where pipelines are heading: AI copilots inside your IDE, your build, and your incident response.

KIRO — GA

Agentic IDE with spec-driven development.

Kiro reached GA in Nov 2025. Spec-driven: describe intent, Kiro generates pipeline, infra, and tests. Kiro CLI brings agents to the terminal., (Kiro autonomous agent)

AWS DEVOPS AGENT — GA

Autonomous on-call engineer for your pipeline.

Autonomous on-call engineer build on Bedrock AgentCore
Correlates CloudWatch, GitHub, CodePipeline signals to identify root causes.
WGU reduced MTTR by 77% — from 2 hrs to 28 mins.

MCP EVERYWHERE

Model Context Protocol as the integration layer.

four managed MCP servers in preview (AWS, EKS ,ECS ,SageMaker.)
Bring-Your-Own-MCP for custom tools.

SECURITY AGENT - GA

AI-assisted design reviews and code analysis.

Performs Ai-driven design reviews, PR level code analysis against OWASP top 10 and On-demand pen-testing .Third frontier agent announced at re:Invent 2025. Guard Duty expanded across EC2 and ECS. Security shifts into the intent phase.

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Key takeaways.

Three things to walk out of this room knowing.

01

CI/CD is a discipline — tools just automate it.

The culture of automated testing and deployment is the real skill. Tools change. The discipline doesn't.

02

CodePipeline is the anchor.

Everything else plugs in. If you remember one name from today, make it CodePipeline.

03

You already understand the bones.

Real production pipelines are this same shape — just with more stages, environments, and guardrails.

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QUESTIONS?

Thank You.

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